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GENERAL WINTER PRECIPITATION FORECASTING

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CATEGORY A. MID-LATITUDE CYCLONE WINTER STORM

*With a winter mid-latitude cyclone the heaviest snow is 100 to 300 kilometers to the left of the cyclone track. The heaviest snow is often in a narrow band. This region has the greatest combination of cold air and uplift. Snow in association with [upper level lows](#) (700 to 500 millibars) tend to be more directly below the low track.

*Watch mid-latitude cyclones that tilt with height. Often upper level low will be displaced hundred(s) of kilometers to west or northwest of surface low. Upper level lows can bring unexpected heavy snow. Model vertical resolution is poorer in upper levels than in lower levels.

Source regions of winter cyclones

- *Colorado lows
- *Canadian lows (generally lighter snow)
- *[CISK](#) (convective instability of the second kind: large latent heat release) lows (Nor-easters)
- *SW US trough (often induced by jet streak)
- *Gulf of Mexico low (often form along pre-existing thermal gradient)

The BIG 7 questions to ask yourself when looking at the models:

1. What will the track of the low be?

- *Rain/snow line depends on track
- *Temperature advections depend on track
- *Use model data from mesoscale and synoptic models

2. How will the low evolve through time (speed, intensity)?

- *This will influence snow amounts

3. How cold will the [PBL](#) temperatures change through time?

- *Determines precipitation type and melting potential
- *Heaviest snow tends to fall in the region where 850 millibar temperatures range from -2 to -6°C (1000 to 500 millibar [thickness](#) between 5,310 and 5,370 m)

4. How much moisture will be advected toward the low?

- *Too much dry air will limit snow amounts
- *High dewpoints in warm sector of cyclone can wrap moisture into cold sector producing more condensation, latent heat release and precipitation amounts
- *1000 to 500 millibar relative humidity needs to be greater than 90% to maximize heavy snowfall

5. How does temperature profile (forecast soundings) evolve around the low?

*The precipitation type may change through time

*[Wet bulb cooling potential](#)

*Forecast soundings

6. What is the potential for dry air to wrap into the low?

*Often air is too dry in cold sector to support snow

7. What mesoscale influences are the synoptic models not going to pick up?

*Heaviest snow falls in a narrow band

*[Orographic](#) influence

Category B: FREEZING DRIZZLE/ FREEZING RAIN/ LIGHT MIX

*May not be directly associated with a mid-latitude cyclone

*Must have polar air mass in place with below zero Celsius temperatures at surface OR wetbulb temperatures below zero at the surface

*Caused by [WAA](#) (aka [Isentropic lifting](#), return flow) from the Gulf or Atlantic. Slang used is overrunning since [differential advection](#) with height is occurring (cP air in PBL, mT air above this layer).

*If return flow is too strong OR temperatures are at freezing, freezing drizzle/rain will change to all rain shortly.

*Can also occur when [shortwaves](#) associated with the subtropical jet transverse Polar air at lower levels. If temperatures are cold enough, light snow will result.

Category C: LAKE EFFECT SNOW (discussed [here](#))

Category D: SLEET

*May occur in transition from rain to snow

*Heavy prolonged sleet events occur with a category B situation but the below zero PBL temperatures are higher in the atmosphere in relation to the surface.

*Can also occur when upper level low transverses a fairly deep layer of PBL polar air and a relatively warm layer caps the PBL

*[Elevated convection](#) may be strong enough to produce thunder sleet or thunder snow

Why do winter forecasts bust so often??

1. For one thing, this is a forecast that is not easy to forget by the public because it will drastically alter their plans. If they see winter precipitation in the forecast, it is a guarantee they will remember if it does not happen. When predicting rain it will be in the liquid form but predicting winter precipitation it can be (snow, wet snow, sleet, [freezing rain](#), just a cold rain, freezing drizzle, a wintry mix, a change from one form of frozen precip to another, etc) which adds to the complexity.

2. [Upward vertical velocity](#) is not strong enough (often [CAA](#) over-powers uplift caused by upper level shortwaves).

3. The air is too dry below 700 mb. Even if precipitation does occur, evaporation can prevent it from reaching the surface. Often dry air wraps into the mid-latitude cyclone, cutting off precipitation in the cold sector.

4. Uplift mechanisms move downstream just as freezing temperatures are advected in. This is especially true in the SW quadrant of the mid-latitude cyclone.

5. Polar air is not cold enough. A one or two degree difference can greatly impact winter forecast. Polar air often modifies due to warm soils. WAA can raise temperatures above freezing just as lifting mechanisms move into place. Also, soil temperatures may be too warm for accumulation.

6. The track of the low was different than expected.

7. Intensity of upper level low was misleading from models. Upper level lows are notorious for their explosive development or rapid decay. Model error in general

8. Not enough moisture. UVV's not strong enough to condense moisture into precipitation.

9. A lack of lifting mechanisms. Especially true for situation in which a low pressure is not present.

Nowcasting:

*Pay close attention to latest satellite and radar when a winter storm is in the developing stages. Radar intensity trends can give an indication whether the storm is strengthening or weakening.

*When observing upper level lows it is best to use satellite data. As soon as the clouds become brighter white in association with the upper level low, the low is deepening. Also use satellite data to see if moisture or dry air is wrapping into the upper level low.

*Satellite and Radar data are best to use in locating where the heaviest winter precipitation is falling.

*Look for mesoscale processes that can increase precipitation intensity (upslope flow, orographic lifting {Rockies, Appalachians, higher elevation regions east of the Great Lakes}, elevated moisture, temperature and moisture mesoscale boundaries).

*Since upper level lows are cold cored, they can produce snow in the winter even when surface temperatures are well above freezing. Locations away from the low are more likely to see rain instead of snow when temperatures are above freezing at the surface.

*Monitor surface temperatures and dewpoints hour by hour for:

- a. temperature advections
- b. moisture advections
- c. wet-bulb cooling
- d. [diabatic heating](#) and cooling (radiational cooling, daytime heating)

*Monitor hour by hour wind speed and direction for:

- a. dry air advection which can increase wet bulb cooling
- b. wind speed helps determine temperature advection

*Read National Weather Service discussions

*Keep in mind data from the synoptic charts

- a. 5,400 gpm thickness
- b. Eta, NGM, AVN model
- c. Skew-T's
- d. Analysis charts
- e. 850 temperatures

